

AOT-GOAT Bias & Performance Audit

Empirical evaluation of recipient-conditioned ambiguity detection and intervention behaviour

59	0.951	0.975	0.963
Evaluation cases	Overall Precision	Overall Recall	Overall F1

1. Objective

This audit evaluates the current AOT-GOAT prototype across overall intervention performance, the ambiguity categories defined by the implementation, and recipient/context conditions that are intended to shape risk. It also examines intervention-rate behaviour and documents limitations revealed by the retained evaluation record. No demographic or sensitive-attribute analysis is introduced because those attributes are not part of the available evaluation basis.

2. Audit Dimensions

The implementation defines five ambiguity categories. The available retained evaluation record does not preserve row-level outputs required to compute defensible per-category precision, recall, or F1, so those subgroup metrics are reported as unavailable rather than estimated.

Risk type	Operational meaning	Audit status
Coreference	Unclear reference to an entity	Represented; per-type metric not retained
Temporal	Ambiguous time expression	Represented; per-type metric not retained
Scope	Unclear range or recipients	Represented; per-type metric not retained
Terminology	Unclear jargon or acronym	Represented; per-type metric not retained
Implicit Context	Meaning depends on unstated shared context	Represented; per-type metric not retained

Recipient/context evidence. The documented profile includes dyadic history, clarification behaviour, terminology usage, recent entities, temporal references, and timezone difference. High/low subgroup thresholds cannot be reconstructed from the retained material, so no arbitrary median or hand-picked split is introduced.

3. Overall Performance

Evaluation set	Cases	Precision	Recall	F1
Dev	41	0.929	1.000	0.963
Holdout	18	1.000	0.929	0.963
Overall	59	0.951	0.975	0.963

Decision-level intervention metrics: Precision 0.969 | Recall 0.969 | F1 0.969. On the 59-case record, this corresponds to the reconstructed decision matrix used on page 2. The split results show a precision-recall trade-off while F1 remains 0.963 in both splits.

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Decision-level audit, context analysis, findings and limitations

4. Performance by Risk Type

Unavailable from the retained evaluation data. The five implementation categories are known, but the raw evaluation rows and category-level prediction export needed for numeric per-risk precision, recall, F1, false-positive rate and false-negative rate were not retained in the accessible record. Reporting values would require inventing or estimating measurements.

5. Performance by Recipient / Context Condition

Context condition	Cases	Precision	Recall	F1	Intervention rate
High vs low shared context	N/A	N/A	N/A	N/A	N/A
Strong vs weak grounding	N/A	N/A	N/A	N/A	N/A
Low vs high clarification history	N/A	N/A	N/A	N/A	N/A

These comparisons are conceptually central to AOT-GOAT, but the accessible evaluation record contains no row-level labels from which valid subgroup rates can be computed. The report therefore separates documented design intent from measurable audit results.

6. Confusion Matrix

	Actual PASS	Actual INTERVENE
Predict PASS	TN = 26	FN = 1
Predict INTERVENE	FP = 1	TP = 31

Reconstruction note. The counts TP=31, TN=26, FP=1 and FN=1 are reconstructed from the 59-case total and the repository-reported decision-level Precision/Recall/F1 of 0.969/0.969/0.969 (rounded to three decimals). The raw confusion-count export was not retained. Derived rates: FPR = $1/(1+26) = 3.7\%$; FNR = $1/(1+31) = 3.1\%$.

7. Intervention-Rate Comparison

Overall predicted INTERVENE **32 / 59** **54.2%**

The 54.2% figure is an overall reconstructed intervention rate, not a high-context versus low-context comparison. No subgroup intervention-rate difference is reported because the required raw grouping fields are unavailable. Consequently, the data support observation of the overall gate behaviour but not a quantitative claim about context-conditioned disparity.

8. Findings + Limitations

- Overall: The retained record reports Precision 0.951, Recall 0.975 and F1 0.963 across 59 synthetic cases.
- Split behaviour: Dev reports 0.929 precision and 1.000 recall; holdout reports 1.000 precision and 0.929 recall; both report F1 0.963.
- Decision errors: The reconstructed matrix contains one false positive and one false negative, corresponding to FPR 3.7% and FNR 3.1%.
- Context audit: Recipient-conditioned evidence is part of the design, but subgroup disparity cannot be quantified from the retained record without row-level context labels and predictions.
- Limitations: Small synthetic, hand-authored/project-created evaluation examples; no retained per-risk/per-context output; results may not generalize to real-world communication; rule/lexicon sensitivity; prototype rather than production-scale system.